

Retail Sales Analytics and Customer Purchasing Behaviour Analysis

CIND820 – Big Data Analytics Project

Milestone 1: Project Design and Strategy

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1. Project Context & Problem Definition

Project Aim

This project focuses on analyzing retail sales data to understand customer purchasing behaviour and sales trends.

Problem

Retail companies collect large amounts of sales transaction data every day. However, businesses may struggle to understand customer purchasing behaviour, product performance, and sales trends without proper analysis.

Who the Project is Meant to Help

This project is meant to help:

- Retail managers
- Marketing teams
- Inventory planners
- Business analysts

Real-World Importance

The project may help businesses understand customer purchasing patterns, identify top-selling products, and improve business decision-making.

2. Dataset Overview & Privacy Review

Dataset

The dataset selected for this project is the Online Retail Dataset from the UCI Machine Learning Repository.

Dataset Link:

<https://archive.ics.uci.edu/dataset/352/online+retail>

Where the Data Came From

The dataset was collected by a UK-based online retail company as part of normal business sales operations.

The data was later published through the UCI Machine Learning Repository for educational and research purposes. I downloaded the dataset from the official UCI public repository.

Dataset Characteristics

The dataset contains:

- Invoice numbers
- Product descriptions
- Quantities purchased
- Unit prices
- Customer IDs
- Invoice dates
- Country information

Dataset Limitations

The dataset has several limitations:

- Data is from only one retail company.
- The data is from 2010–2011.
- Some customer IDs are missing.
- Some invoices represent cancellations or return.
- Some records may require cleaning and validation.

Privacy and Ethics Review

The dataset does not contain names, phone numbers, addresses, or email addresses.

The project will use the dataset only for educational purposes. No attempt will be made to identify customers.

Possible risks include:

- Limited representation because data is from one company only
- Possible business bias
- Misinterpretation of customer purchasing patterns

The dataset is publicly available for educational and research use through the UCI Machine Learning Repository.

3. Research Questions and Impact Hypothesis (Project Goals)

Research Questions

1. Which products generate the highest sales revenue?
2. Which countries and customer groups contribute the most to total revenue?
3. What monthly or seasonal sales trends can be identified from the retail transaction data?

Why These Questions Matter

These questions help businesses understand customer purchasing behaviour, sales performance, and product demand.

Impact Hypothesis

A useful result would identify top-selling products, important customer groups, and seasonal sales trends. These findings may help businesses improve inventory planning, marketing strategies, and sales decisions.

4. Project Workflow & Analysis Plan

The Project will follow a step-by-step data analysis process.

Step 1: Data Collection-:

The dataset will be downloaded from the UCI Machine Learning Repository and imported into the analytical environment.

Step 2: Data Understanding and Initial Audit-:

The dataset structure, variables, missing values, invoice types, and data distributions will be reviewed to understand the overall quality and characteristics of the retail transaction data.

Step 3: Data Cleaning and Preparation-:

Data cleaning activities may include:

- Removing or handling missing values
- Identifying cancellation transactions
- Cleaning invalid quantities or prices
- Creating calculated revenue fields
- Formatting date variables
- Preparing data for analysis and visualization

Step 4: Exploratory Data Analysis (EDA)-:

Exploratory analysis will be conducted using:

- Descriptive statistics
- Revenue analysis
- Product performance analysis
- Country-wise sales analysis
- Customer purchasing behaviour analysis
- Time-series and seasonal trend analysis
- Data visualizations and charts

Step 5: Advanced Analytics and Business Insights-:

Additional analysis may include:

- Customer segmentation
- Revenue contribution analysis
- Trend identification
- Basic forecasting techniques
- Correlation analysis

Step 6: Evaluation and Interpretation-:

The project findings will be evaluated and interpreted to determine:

- Major sales contributors
- Seasonal purchasing behaviour
- High-value customer groups
- Product demand patterns
- Business improvement opportunities

Step 7: Visualization and Reporting-:

Results will be presented using professional charts, dashboards, and visualizations created using Python libraries and Power BI where appropriate.

5.GenAI Declaration

Generative AI tools were used for brainstorming and wording support during the preparation of this project.

All analysis, decisions, and final content were completed and reviewed manually.

6. References

Chen, D. (2015). Online Retail Dataset. UCI Machine Learning Repository. Retrieved from <https://archive.ics.uci.edu/dataset/352/online+retail>

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